



PHILOMATH NETWORKS

**PHILOMATH
NETWORKS PVT. LTD**



**6-Days Hands-On
Workshop
on**

“Machine Learning Certification Course”

**Prepare yourself for the Machine
Learning Certification with us!**

**Redefining Learning
Beyond Boundaries**



**Phone Number
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About Philomath Networks



At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!



Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.

End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.

Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course

Overview



Course Objective

Machine Learning Certification Training course builds a strong foundation in Machine Learning (ML) and Artificial Intelligence (AI) by covering key concepts, algorithms, and real-world applications. Participants will explore supervised, unsupervised, and deep learning, along with data preprocessing, model evaluation, and hands-on implementation using Python, Scikit-Learn, TensorFlow. A Titanic Survival Prediction project reinforces learning. The course also covers AI ethics, trends, and career opportunities, equipping learners to develop, optimize, and deploy ML models for AI-driven careers.

Course Outcome

✓ Understand AI & ML Concepts

✓ Develop Deep Learning Models

✓ Preprocess & Engineer Data

✓ Build & Evaluate ML Models

✓ Implement ML Algorithms

✓ Explore AI Ethics & Career Paths

Training

Modules

The Machine Learning Certification Training Course covers six modules, blending theory with practical applications. It begins with Introduction to AI & ML, exploring key concepts, AI vs. ML vs. Deep Learning, real-world applications, and ethics. Machine Learning Fundamentals introduces key algorithms, learning types, dataset preparation, and evaluation metrics. Data Preprocessing & Feature Engineering focuses on data cleaning, handling missing values, feature scaling, and dimensionality reduction to improve model performance.

Module 1: Introduction to AI & Machine Learning

- Define AI & ML, their significance, and impact
- Compare AI, ML, and Deep Learning with real-world examples
- Explore major AI & ML applications across industries
- Discuss ethical considerations and responsible AI practices
- Understand the basics of data-driven decision-making
- Recognize career opportunities and growth in AI & ML

Module 2: Machine Learning Fundamentals

- Learn types of ML: Supervised, Unsupervised, Reinforcement Learning
- Understand key ML algorithms and their applications
- Explore the role of training, validation, and testing data
- Learn model evaluation metrics: Accuracy, Precision, Recall, F1-Score
- Study bias-variance tradeoff and overfitting vs. underfitting
- Discuss challenges in implementing ML models

Module 3: Data Preprocessing & Feature Engineering

- Understand the importance of data quality in ML
- Learn techniques for handling missing values & outliers
- Apply feature scaling and normalization methods
- Explore feature selection techniques to enhance model performance
- Implement dimensionality reduction (PCA, LDA)
- Perform exploratory data analysis (EDA) for insights



Training Modules

The later modules focus on applying ML techniques, covering Supervised & Unsupervised Learning with regression, classification, and clustering models. Deep Learning & Neural Networks introduces TensorFlow, Keras, CNNs, and RNNs. The final Hands-on Project & AI Ethics module includes the Titanic Survival Prediction project, model deployment, and AI ethics, ensuring learners gain practical skills and industry readiness. Participants will also explore AI fairness, explainability, and emerging trends shaping the future of ML. This structured approach equips learners with essential skills for AI-driven careers.

Module 4: Supervised & Unsupervised Learning

- Explore regression models (Linear, Polynomial, Logistic)
- Study classification models (Decision Trees, Random Forest, SVM, k-NN)
- Understand hyperparameter tuning for performance optimization
- Implement clustering techniques (K-Means, Hierarchical Clustering)
- Learn Principal Component Analysis (PCA) for dimensionality reduction
- Compare supervised and unsupervised learning applications



Module 5: Deep Learning & Neural Networks

- Understand the basics of neural networks and deep learning
- Learn about activation functions and backpropagation
- Explore TensorFlow framework for DL applications
- Study Convolutional Neural Networks (CNNs) for image processing
- Implement Recurrent Neural Networks (RNNs) for sequential data
- Discuss real-world applications of deep learning



Module 6: Hands-on ML, AI Ethics, Future Trends & Certification Exam

- Implement Titanic Survival Prediction using Python & Scikit-Learn
- Train, test, and evaluate an ML classification model
- Understand real-world model deployment strategies
- Address bias, fairness, and responsible AI practices
- Explore Explainable AI (XAI), emerging ML advancements, and career paths
- Assess knowledge through a final exam covering key AI & ML concepts



Training

Schedule & Format



The Machine Learning Certification Training Course offers a comprehensive learning experience through six interactive modules over six days/weeks. It combines theoretical foundations with hands-on practice, covering data preprocessing, model development, evaluation, and deployment. Participants will work with real datasets, implement ML algorithms, and apply best practices to solve real-world problems. Through practical projects and case studies, they will gain essential machine learning skills, understand AI ethics, and explore industry trends to make data-driven decisions.

Day	Module	Focus Area	Duration
Day 1	Introduction to AI & Machine Learning	ML Basics, Applications	5hrs
Day 2	Machine Learning Fundamentals	Cleaning, Scaling	5hrs
Day 3	Data Preprocessing & Feature Engineering	Algorithms, Models	5hrs
Day 4	Supervised & Unsupervised Learning	Training, Metrics	5hrs
Day 5	Deep Learning & Neural Networks	Neural Nets, CNN	5hrs
Day 6	Hands-on ML, AI Ethics, Future Trends & Certification Exam	Project, Deployment	5hrs

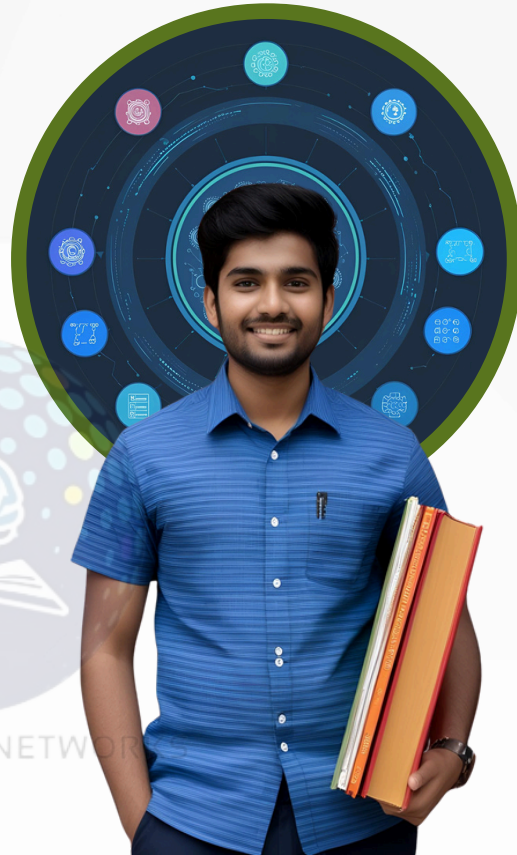
The workshop combines live demonstrations, guided projects, and collaborative activities to enhance learning and retention. Participants will tackle industry-relevant challenges, simulating real-world machine learning applications. The interactive format fosters discussions, troubleshooting, and peer learning. By the end, attendees will gain a strong foundation in machine learning, enabling them to build, evaluate, and deploy ML models. Whether professionals enhancing their AI skills or beginners entering the field, this workshop equips them with essential tools for success in a data-driven world.

How to Get Involved



Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact Information

For any questions about the training program, please reach out to:

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